

"Do not write anything on question paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules"

Roll No. 20UMEC1121

BE IV (M)

3

Internal Comb. Engines

FINAL B.E. EXAMINATION, 2023
MECHANICAL ENGINEERING
(VII SEMESTER CBCS)

ME 402 A : INTERNAL COMBUSTION ENGINES

Time – Three Hours

Maximum Marks – 80

- Note: (i) Attempt any **Five** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1. (a) Bring out clearly the process of combustion in SI engines and also explain the various Stages of combustion. 10
(b) A diesel engine is used in a truck requiring 80 kW. The mechanical efficiency of the engine is 80%. The brake fuel consumption of the engine is 258 gm per KW-hr. A design Improvement is made which reduces the engine friction by 3.7 kW. Assuming the indicated thermal efficiency remains the same, calculate
(i) the new mechanical efficiency
(ii) the new bsfc (brake specific fuel consumption).
(iii) the saving in fuel per hour 6
2. (a) Explain the phenomenon of diesel knock. Compare it with the phenomenon of detonation in S.I. Engines. 8
(b) Explain with figures the various types of Combustion chambers used in CI engines. 8
3. (a) Enlist and discuss the important properties of a lubricant which affect engine performance. 8
(b) Why over-cooling in engine is harmful? Describe force circulation system with a neat sketch. 8

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Cont.

4.

(a)

Explain the scavenging process in two-stroke engine. What is meant by blowdown? 10

(b)

For a 2-cylinder two Stroke I.C. Engine the following data are given: - engine speed: 4000 rpm; volumetric efficiency: 77% mechanical efficiency: 75%; fuel consumption: 10 liter/hr Specific gravity of fuel 0.73; calorific value of fuel 43932 KJ/Kg; Air fuel ratio: 18:1 piston Speed: 600 m/min; Indicated mean effective pressure: 5 atm. Find out

(i) engine dimensions

(ii) brake power

(iii) thermal efficiency

Assume Standard pressure and temperature. 6

5.

(a)

What are catalytic converter? How do they help in reducing HC, CO and NO_x emission? 10

(b)

What are the causes and problems of exhaust emissions? 6

6.

(a)

Explain MPFI System with a neat Sketch. 8

(b)

The air flow to a four cylinder, four stroke petrol engine is measured by means of a 7.5 cm diameter Sharp-edged orifice $C_d=0.6$. During a test on the engine following data were recorded.

Bore = 11 cm; Stroke = 13 cm; engine speed=2250 rpm; brake power=36 kW; fuel Consumption=10.5 kg/hr; calorific value of fuel=42,000 KJ/Kg; pressure drop across the orifice = 4.1 cm of water. Atmospheric temperature and pressure are 15°C and 1.013 bar. Calculate

i. Thermal efficiency on b.p. basis

ii. Brake mean effective pressure

iii. Volumetric efficiency based on free air condition. 8

7.

(a)

Write Short notes on the following (ANY FOUR). 4x4=16

(i) HUCR Dope.

(ii) Dual fuel and multifuel engine

(iii) CRDI System

(iv) Supercharging

(v) Free piston engine

(vi) Rotary engine

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Roll No. 210MECC114.1

BE IV (ME)

3

IC Engine

FINAL B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

(VII SEMESTER CBCS)

ME-402 A: INTERNAL COMBUSTION ENGINES

Time - Three Hours

Maximum Marks - 80

- Note:- (i) Attempt any **Five** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1. (a) Describe the phenomenon of detonation or knocking in S.I. engine. On what factors does detonation depend? 10
(b) With a neat sketch explain working of Multi Point Fuel Injection System (MPFI). 6
2. (a) Explain the process of combustion in CI Engine compare it with the combustion of SI Engine. 10
(b) What are the three methods of generating swirl in CI engine combustion chambers? 6
3. A test on a single cylinder, four stroke diesel engine having bore 18cm and stroke 36 cm yielding the following results speed 285 rev/min; brake torque 393 Nm. indicated mean effective pressure 7.2 bar; fuel consumption 3.5 kg/hr;

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cooling water flow 4.5 kg/min; cooling water temp' rise 36°C; air - fuel ratio by mass 25; exhaust gas temp'; 415°C; barometric pr. 1.013 bar; room temp' 21°C.

The fuel has a calorific value of 45200 kJ/kg. Determine.

- (a) The indicated thermal efficiency; 38.65% 5
 (b) The volumetric efficiency based on atmospheric conditions. 3.77×10^{-3} 5

Draw up a heat balance in terms of kJ/min explaining clearly the content of such term.

Take $R = 0.287 \text{ KJ/Kg K}$

$C = 1.005 \text{ KJ/kg K}$ for exhaust gases. 6

4. (a) Explain with suitable sketches the following scavenging system. 12

- (i) Uniflow Scavenging
 (ii) Cross flow Scavenging
 (iii) Loop or reverse scavenging

- (b) Explain the various scavenging pumps used in two stroke engine. 4

5. (a) Explain the working of ionisation detector used to measure HC- emission in the exhaust. 8

- (b) Discuss the emissions from CI engines. What are the factors which effect CI engine emissions? 8

6. (a) Draw a neat diagram of thermostatic controlled water cooling system and explain its working. 8

- (b) What is meant by dry and wet sump lubrication? Where dry sump is preferred and why? 8

7. Write short notes 4X4

- (i) Quel fuel and multi fuel engines
 (ii) HUCR dopes
 (iii) Types of super charging
 (iv) Free piston engines.